DES432: Statistics and Data Modeling

Week 14: Data Simulation, Bootstrapping, and Monte Carlo Methods

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Course Progression: Where We Are Now

Estimation $\rightarrow$ Testing $\rightarrow$ Regression $\rightarrow$ Diagnostics $\rightarrow$ Evaluation $\rightarrow$ Simulation

Weeks 9–14 at a glance:

- Week 9: Estimation and confidence intervals
- Week 10: Hypothesis testing
- Week 11: Correlation and simple regression
- Week 12: Multiple regression and diagnostics
- Week 13: Model evaluation and comparison
- Week 14: Simulation-based inference and advanced applications

Key Idea

We now move from analytical formulas toward simulation-based reasoning.

Conceptual Shift from Week 13 to Week 14

Week 13 Question
Which model is better?

Week 13 focused on:

- goodness-of-fit,
- prediction error,
- overfitting and underfitting,
- model comparison.

Week 14 Question
What can we do when formulas are difficult, assumptions are unclear, or uncertainty is hard to compute directly?

Week 14 goes one step further:

- simulate data,
- resample from observed data,
- approximate uncertainty computationally.

Why This Topic Matters

In statistics, many important questions are about variability:

- How much might the sample mean change?
- How uncertain is a regression coefficient?
- How stable is a prediction error measure?
- What if the assumptions behind a formula are doubtful?

Simulation gives a practical way to explore such questions.

Key Idea

A single dataset gives one answer. Simulation helps us see many possible answers.

Learning Objectives

After this session, you should be able to:

1. explain the idea of data simulation,
2. describe how bootstrapping works,
3. explain the basic logic of Monte Carlo methods,
4. interpret simulation-based uncertainty,
5. connect simulation ideas to model evaluation and Project 2.

Common Pitfall

Students sometimes think simulation is only a programming trick. In fact, it is a statistical method for approximating uncertainty and studying repeated-sample behavior.

A Little History

Simulation-based thinking became especially important when statisticians and scientists wanted to study problems that were too difficult for simple formulas.

Very brief historical picture:

- Probability and random reasoning developed much earlier in classical mathematics.
- With the growth of computers in the 20th century, repeated numerical experiments became practical.
- Monte Carlo methods became widely known in mid-20th-century scientific computing.
- Bootstrapping became a major modern statistical tool for resampling-based inference.

Key Idea

The rise of computing changed statistics from only formula-based analysis to computation-based inference.

Why the Name "Monte Carlo"?

The name "Monte Carlo" was given by **Nicholas Metropolis**, after **Monte Carlo, Monaco**, which is famous for casinos and games of chance.

The method was named this way because it uses:

- randomness,
- repeated trials,
- chance-based outcomes

to approximate quantities that may be difficult to calculate directly.

Historical Note

The basic idea was first conceived by **Stanislaw Ulam** in the **mid-1940s** (around 1946), and later developed further with **John von Neumann**, **Nicholas Metropolis**, and others in the late 1940s.

Example

Instead of deriving a complicated probability exactly, we simulate the experiment many times and estimate the probability from the proportion of successes.

From Formula-Based to Simulation-Based Inference

Classical methods often rely on:

- Normality assumptions,
- large-sample approximations,
- closed-form standard error formulas.

Simulation-based methods rely on:

- repeated random generation,
- resampling from data,
- empirical distributions of statistics.

Comparison

Classical inference is often algebra-first. Simulation-based inference is often computation-first.

Why Do We Need Simulation?

In practice:

- the sample may be small,
- the population distribution may be unknown,
- the statistic may be complicated,
- the model may be too complex for direct calculation.

Simulation helps when direct formulas are hard, unavailable, or less trustworthy.

Key Idea

Simulation studies variability by imitating the random process many times.

What Is Data Simulation?

Data simulation means generating artificial observations from a specified random process.

Examples:

- simulate coin flips,
- simulate exam scores,
- simulate repeated samples from a population,
- simulate regression errors.

Why do this?

- to understand randomness,
- to study sampling variability,
- to estimate uncertainty,
- to test how a method behaves.

General Simulation Workflow

A simulation study often follows these steps:

1. Define the process or model.
2. Generate one sample.
3. Compute a statistic of interest.
4. Repeat many times.
5. Study the distribution of the results.

Example

If we want to understand how sample means vary, we can repeatedly simulate samples and compute the mean each time.

One Outcome vs Many Outcomes

A single observed dataset is only one realization of a random process.

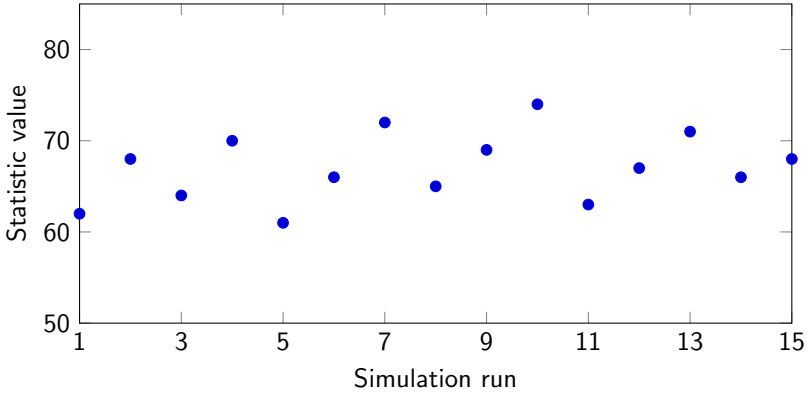
If we could repeat the data collection many times:

- the statistic would change,
- the estimate would fluctuate,
- uncertainty would become visible.

Key Idea

Statistics is not only about the observed answer. It is about the range of answers that could plausibly have occurred.

Visual Intuition: Repeated Outcomes



A single run gives one value. Many runs reveal variability.

Connection to Sampling Distributions

Earlier in the course, we discussed the idea of a sampling distribution:

- if we repeatedly draw samples,
- the statistic changes from sample to sample.

Simulation helps us make this idea concrete.

Key Idea

Simulation turns the abstract idea of a sampling distribution into something we can visualize.

Example: Repeated Sample Means

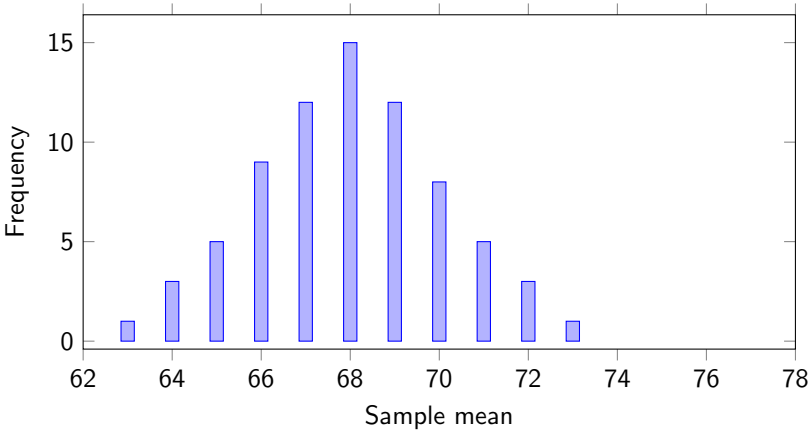
Suppose a population has mean μ.
If we repeatedly draw samples of size n and compute the sample mean $\bar{x}$:

- each sample mean is slightly different,
- the collection of these means forms a distribution,
- this distribution tells us how variable $\bar{x}$ is.

Interpretation

If the sample means cluster tightly, the estimator is relatively stable. If they spread out widely, the estimator is more uncertain.

Visual: Sampling Distribution Shape



Key Idea

Repeated sampling creates a distribution of the statistic, not just one number.

Bootstrapping: Main Idea

Bootstrapping is a resampling method.
Instead of sampling from the population directly, we:

- use the observed sample,
- sample from it **with replacement**,
- compute the statistic repeatedly.

This gives an approximate distribution of the statistic.

Key Idea

Bootstrapping treats the sample as a stand-in for the unknown population.

What Does "With Replacement" Mean?

Suppose the observed data are:

$$(52,\ 60,\ 63,\ 68,\ 72)$$

A bootstrap sample of size 5 might be:

$$(60,\ 60,\ 72,\ 52,\ 68)$$

Important features:

- some values may appear more than once,
- some original values may not appear at all.

Common Pitfall

Students sometimes resample without replacement. That is not bootstrapping.

Bootstrap Procedure

To bootstrap a statistic:

1. Start with the observed dataset of size n.
2. Draw a bootstrap sample of size n with replacement.
3. Compute the statistic of interest.
4. Repeat many times.
5. Study the bootstrap distribution.

Possible statistics

Mean, median, regression slope, correlation, prediction error

Visual: Bootstrap Resampling

Original	70	72	75	78	80
Bootstrap	72	80	72	75	70

Observation

The bootstrap sample has the same size as the original sample, but repeated values can appear.

Example: Bootstrap the Sample Mean

Observed scores:

$$70,\ 72,\ 75,\ 78,\ 80$$

Sample mean:

$$\bar{x} = \frac{70 + 72 + 75 + 78 + 80}{5} = 75$$

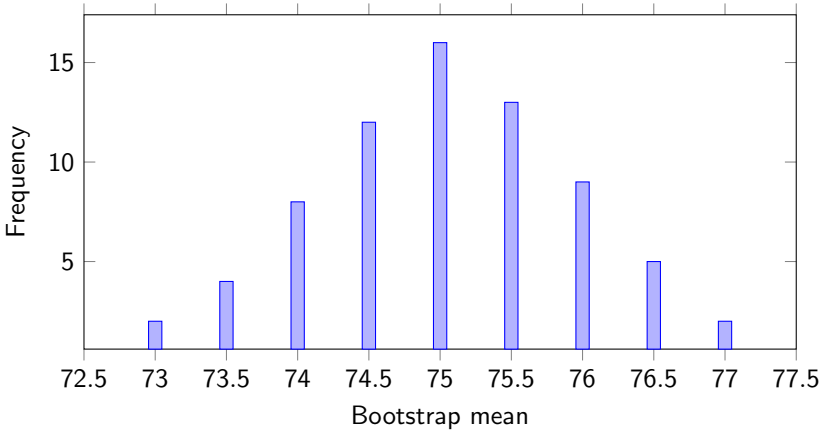
Now create many bootstrap samples and compute the mean each time.
Possible bootstrap means:

$$74.2,\ 76.0,\ 75.6,\ 73.8,\ 77.4, \ldots$$

Interpretation

The spread of these bootstrap means tells us how much the sample mean might vary from sample to sample.

Visual: Bootstrap Distribution of the Mean



Key Idea

The bootstrap distribution approximates the sampling distribution using only the observed data.

What Can We Learn from a Bootstrap Distribution?

From the bootstrap distribution, we can examine:

- the center: approximate estimate,
- the spread: approximate uncertainty,
- the shape: symmetry or skewness,
- percentile-based confidence intervals.

Example

A narrow bootstrap distribution suggests the estimate is relatively stable. A wide one suggests more uncertainty.

Bootstrap Confidence Interval: Basic Idea

A simple bootstrap confidence interval can be built from percentiles.
If we generate many bootstrap statistics:

$$\theta_1^*, \theta_2^*, \ldots, \theta_B^*$$

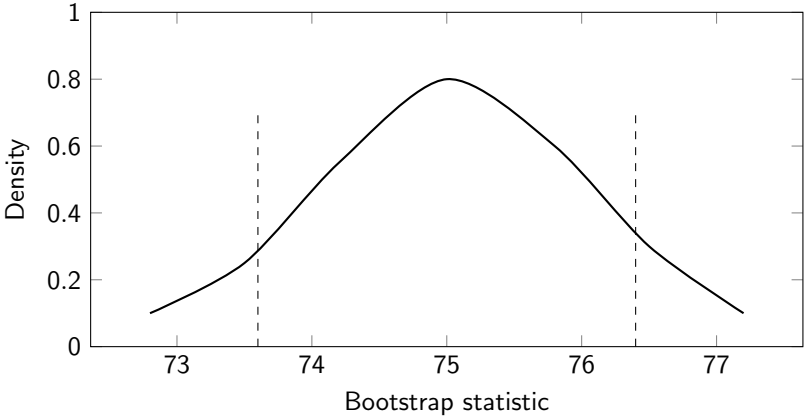
then a rough 95% bootstrap interval can be taken from:

- the 2.5th percentile,
- the 97.5th percentile.

Key Idea

The interval comes from the simulated distribution, not from a Normal formula.

Visual: Bootstrap CI from Percentiles



Approximate 95% bootstrap interval:

$$[73.6,\ 76.4]$$

Classical CI vs Bootstrap CI

Classical confidence intervals often use:

- standard error formulas,
- Normal or t approximations.

Bootstrap confidence intervals use:

- repeated resampling,
- empirical percentiles from the bootstrap distribution.

Comparison

Classical methods are formula-based. Bootstrap methods are simulation-based.

When Is Bootstrapping Useful?

Bootstrapping is especially useful when:

- the sample size is modest,
- the sampling distribution is hard to derive,
- the statistic is complicated,
- we want a data-driven approximation of uncertainty.

Common Pitfall

Bootstrapping is helpful, but it does not fix poor data quality or serious dependence problems automatically.

Monte Carlo Methods: Main Idea

Monte Carlo methods use repeated random simulation to estimate a quantity of interest.
Typical quantities:

- probabilities,
- expected values,
- integrals,
- prediction performance under uncertainty.

Key Idea

If direct calculation is difficult, repeated simulation can approximate the answer.

Monte Carlo Estimation of a Probability

Suppose an event occurs in some proportion of repeated simulations.
If we run the simulation N times, then:

$$\hat{p} = \frac{\text{number of times the event occurs}}{N}$$

This gives a Monte Carlo estimate of the probability.

Example

Simulate 1000 experiments and observe that the event occurs 284 times. Then:

$$\hat{p} = \frac{284}{1000} = 0.284$$

Example: Monte Carlo for Coin Flips

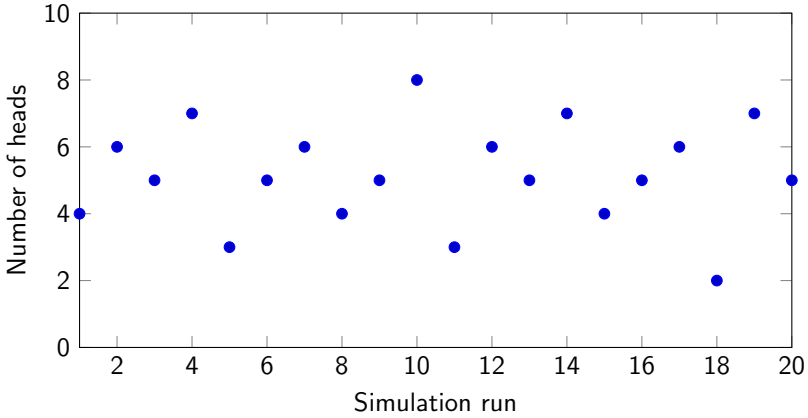
Question: What is the probability of getting at least 7 heads in 10 fair coin flips?
Monte Carlo approach:

1. Simulate 10 fair flips.
2. Count the number of heads.
3. Record whether the count is at least 7.
4. Repeat many times.
5. Estimate the probability by the proportion of successes.

Key Idea

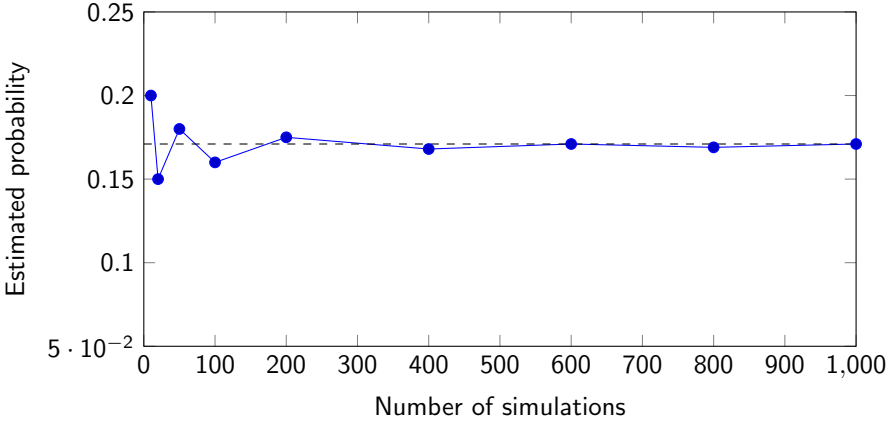
This method approximates a probability experimentally.

Visual: Simulated Number of Heads



Different runs produce different outcomes because the process is random.

Visual: Monte Carlo Estimate Becoming Stable



Key Idea

As the number of simulations increases, the estimate usually becomes more stable.

Simulation Size and Approximation Quality

In general:

- very few simulations may give a noisy estimate,
- more simulations usually reduce Monte Carlo error,
- computational cost also increases.

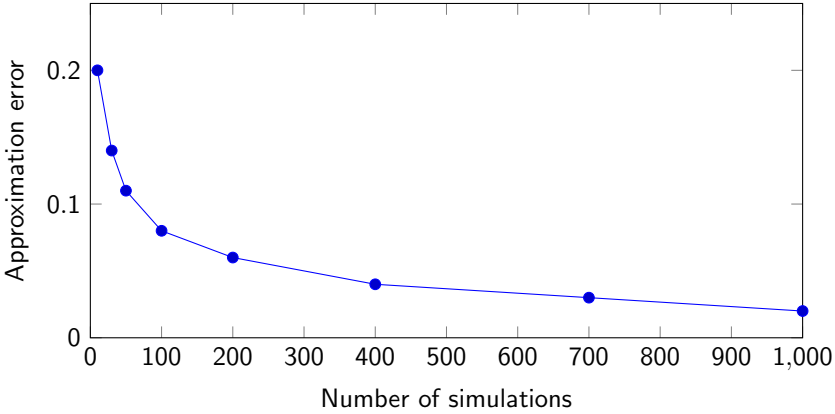
Practical idea

A few dozen simulations are usually too small for stable inference. Hundreds or thousands are more common.

Common Pitfall

A simulated estimate that looks precise may still be unstable if the number of simulations is too small.

Visual: Error Decreases with More Simulations



Key Idea

More simulations do not guarantee perfection, but they often improve approximation.

Bootstrapping vs Monte Carlo

Method	Main source of randomness	Main purpose
Bootstrapping	observed data	estimate uncertainty
Monte Carlo	assumed model/process	approximate probability or behavior

Key Distinction

Bootstrapping resamples from data. Monte Carlo simulates from a model or experiment.

Simulation in Regression Context

Simulation ideas can be used in regression to study:

- uncertainty of slope estimates,
- variability of predictions,
- model performance across repeated samples,
- robustness of conclusions.

Example

Bootstrap the regression slope by repeatedly resampling the observed dataset and refitting the regression model.

Example: Bootstrapping a Regression Slope

Suppose we fit:

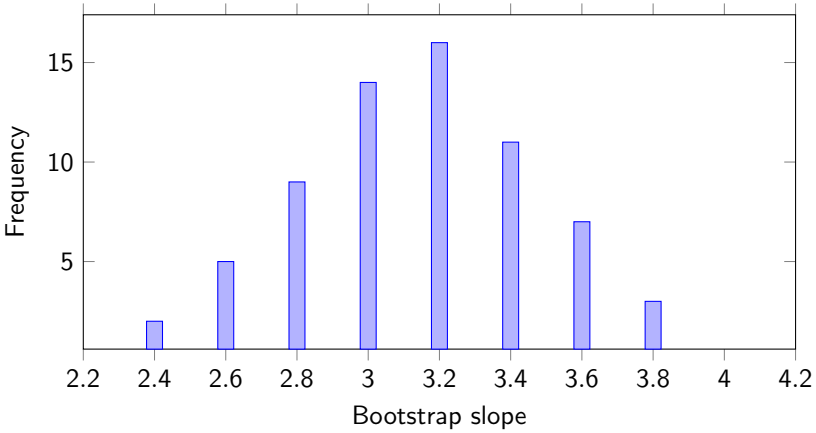
$$\hat{y} = b_0 + b_1 x$$

We want to understand the uncertainty in b_1.
Bootstrap strategy:

1. resample the data with replacement,
2. fit the regression again,
3. record the new slope,
4. repeat many times.

This produces a bootstrap distribution of slope estimates.

Visual: Bootstrap Distribution of Slope



Interpretation

The center suggests the typical slope estimate. The spread reflects uncertainty in the slope.

Connection to Model Evaluation

Week 13 focused on:

- MSE,
- RMSE,
- Adjusted R^2,
- balancing fit and complexity.

Week 14 adds:

- how stable are those results?
- how much might the estimates vary?
- how sensitive is the model to sampling randomness?

Key Idea

Simulation does not replace evaluation. It deepens our understanding of the uncertainty behind the evaluation.

Simulation for Uncertainty Analysis

A major use of simulation is uncertainty analysis.
Typical questions:

- How uncertain is the estimate?
- How much can the prediction change?
- How sensitive is the conclusion to random variation?

Examples

Bootstrap uncertainty of a coefficient, simulate future outcomes under random noise, examine the range of predicted values.

When Should We Use Simulation?

Simulation is useful when:

- a closed-form solution is difficult,
- we want to visualize variability,
- analytical assumptions are doubtful,
- we want to study repeated-sample behavior.

Examples

Bootstrap a confidence interval for a median, simulate a probability in a complex experiment, study prediction error under random noise.

When Should We Be Careful?

Simulation is powerful, but caution is still necessary:

- bad data still lead to weak conclusions,
- dependence problems can distort resampling results,
- unrealistic simulation models produce misleading answers,
- interpretation is still required.

Common Pitfall

A complicated simulation may look impressive but still be based on poor assumptions.

Common Student Mistakes

Be careful with these:

- confusing resampling with ordinary repeated sampling,
- resampling without replacement,
- using too few simulations,
- reporting only a graph without interpretation,
- forgetting what quantity is being estimated.

Rule

Statistical skill = correct method + correct interpretation + honest limitations

Classroom Workflow for a Simulation Problem

A practical workflow:

1. Identify the quantity of interest.
2. Decide whether to simulate from data or from a model.
3. Repeat the process many times.
4. Summarize the simulated results.
5. Interpret the result in context.

DES432 Mindset

Do not stop at the code or the graph. Explain what the simulation means.

Mini Case Study

Suppose we have a small dataset of student scores and want to report uncertainty for the mean. Two possible approaches:

- classical confidence interval using a formula,
- bootstrap confidence interval using resampling.

Questions:

- Are the answers similar?
- Does the sample size seem large enough for the formula-based approximation?
- Does the bootstrap distribution look symmetric or skewed?

In-Class Activity

Use a small numerical dataset.
Tasks:

1. Compute the sample mean.
2. Create several bootstrap samples.
3. Compute the bootstrap mean for each sample.
4. Sketch or interpret the bootstrap distribution.
5. Explain what the spread tells us.

Extension

If time allows, compare a classical confidence interval to a bootstrap confidence interval.

Discussion Questions

After doing a small bootstrap or Monte Carlo example, ask:

- What is the quantity being estimated?
- What creates the variability in the result?
- Is the distribution narrow or wide?
- Would more simulations help?
- What assumptions are still present?

Key Idea

Simulation answers are meaningful only when we know what process is being repeated and what the output represents.

Project 2 Connection

A stronger Project 2 analysis may include:

- a fitted model,
- evaluation metrics,
- discussion of assumptions,
- a simulation-based extension.

Possible extensions:

- bootstrap confidence interval for a coefficient,
- simulation of prediction error,
- Monte Carlo study of uncertainty under repeated sampling.

Key Message

Simulation is a natural extension when you want to go beyond a single fitted model and discuss uncertainty more deeply.

How This Fits the Big Picture of the Course

Across Weeks 9–14, the course has developed this logic:

- Week 9: quantify uncertainty in estimation,
- Week 10: test claims statistically,
- Week 11: model relationships,
- Week 12: diagnose model validity,
- Week 13: compare models,
- Week 14: study uncertainty by simulation.

Big Picture

Simulation-based inference extends the earlier ideas rather than replacing them.

Summary and Closing

Today's main ideas:

- Data simulation studies variability through repeated random generation.
- Bootstrapping resamples from observed data.
- Monte Carlo methods simulate from a random process or model.
- Simulation-based inference helps when formulas are difficult or assumptions are unclear.
- These ideas extend model evaluation by showing uncertainty more directly.

Final Takeaway

Simulation-based reasoning helps us move from one observed result to a broader understanding of possible results.

Preview of Next Week

Next steps:

- final project presentations,
- integration of course concepts,
- connecting probability, inference, modeling, evaluation, simulation, and decision-making.

$$\text{Probability} \to \text{Inference} \to \text{Modeling} \to \text{Evaluation} \to \text{Simulation} \to \text{Decision}$$